Literature Status for arXiv:math/0201144: The Little Hölder Space Is an M-Ideal

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Status

This is a `literature_already_answered` packet. Berninger and Werner ask whether the pointed little Hölder space $H^0_\alpha = \mathrm{lip}_0([0,1], |\cdot|^\alpha)$ is an M-ideal in $H_\alpha = \mathrm{Lip}_0([0,1], |\cdot|^\alpha)$, for $0 < \alpha < 1$ [1]. Kalton's 2004 Theorems 6.1 and 6.6 and the remark immediately after Theorem 6.6 answer this affirmatively, in fact for every compact metric space equipped with a nontrivial gauge [2].

Source Question

On page 4 of the journal article (the final paragraph of the introduction), Berninger and Werner state that it remains open whether H^0_α is an M-ideal in H_α. Section 4 revisits the same question and asks whether their "almond function" might provide a counterexample. Their notation is pointed: the functions vanish at the base point 0, and the norm is the Hölder/Lipschitz seminorm, which is then a norm.

Supporting Answer

Let M be compact and let ω be a nontrivial gauge. Kalton's Theorem 6.1 states

$$\mathrm{lip}_\omega(M)^* = \mathcal{F}_\omega(M), \qquad \mathrm{lip}_\omega(M)^{**} = \mathrm{Lip}_\omega(M).$$

Thus $\mathrm{lip}_\omega(M)$ is the canonical predual of its Lipschitz-free space. Theorem 6.6 proves that, for every compact metric space M and every $\varepsilon > 0$, $\mathrm{lip}(M)$ is $(1+\varepsilon)$-isometric to a subspace of c_0. The remark immediately following the theorem records the relevant consequence: whenever M is compact and $\mathrm{lip}(M)$ is a predual of $\mathcal{F}(M)$, it is an M-ideal in $\mathrm{Lip}(M)$.

Apply this with $M = [0,1]$ and $\omega(t) = t^\alpha$. This is a nontrivial gauge for $0 < \alpha < 1$, so Theorem 6.1 supplies the predual hypothesis and the post-Theorem 6.6 remark gives

$$H^0_\alpha = \mathrm{lip}_\omega([0,1]) \quad \text{is an } M\text{-ideal in} \quad H_\alpha = \mathrm{Lip}_\omega([0,1]).$$

Hence the answer to the source question is *yes*.

Scope and provenance

The answer is stronger than the interval problem: the same implication holds for every compact metric M with a nontrivial gauge. The identification is not a new proof; it is a direct application explicitly signposted by Kalton's remark. A later thesis and conference abstract by Niglas independently name Kalton's Theorem 6.6 as the resolution of the Berninger–Werner problem.

References

[1] H. Berninger and D. Werner, *Lipschitz spaces and M-ideals*, Extracta Math. **18** (2003), 33–56; arXiv:math/0201144.

[2] N. J. Kalton, *Spaces of Lipschitz and Hölder functions and their applications*, Collect. Math. **55** (2004), 171–217, doi:10.1344/CM.V55I2.4055.